

Data Warehouse Project Report

1. Introduction

The OLTP movie rental database supports daily operations such as customer registration, film management, rentals, inventory tracking, and payments. However, it is not optimized for analytical reporting.

This project redesigns the operational schema into a dimensional data warehouse model to support business intelligence and analytical queries related to rentals, revenue, customers, films, inventory, and store performance.

2. Dimensional Model Design

2.1 Selected Business Processes

The warehouse is designed around three main business processes:

1. Film rentals
2. Payments
3. Inventory management

2.2 Fact Tables

FactRental

Grain: One row per rental transaction.

Measures:

- rental_duration
- late_return_flag

Dimensions:

- DimCustomer
- DimFilm
- DimStore
- DimStaff

- DimDate

FactPayment

Grain: One row per payment transaction.

Measures:

- amount

Dimensions:

- DimCustomer
- DimFilm
- DimStore
- DimStaff
- DimDate

FactInventory

Grain: One row per inventory item per store.

Measures:

- availability_flag
- rental_frequency

Dimensions:

- DimFilm
- DimStore

2.3 Dimension Tables

DimCustomer

Stores customer details such as name, email, city, and country.

DimFilm

Stores film information such as title, release year, rating, length, and language.

DimStore

Stores store location information.

DimStaff

Stores staff information and associated store.

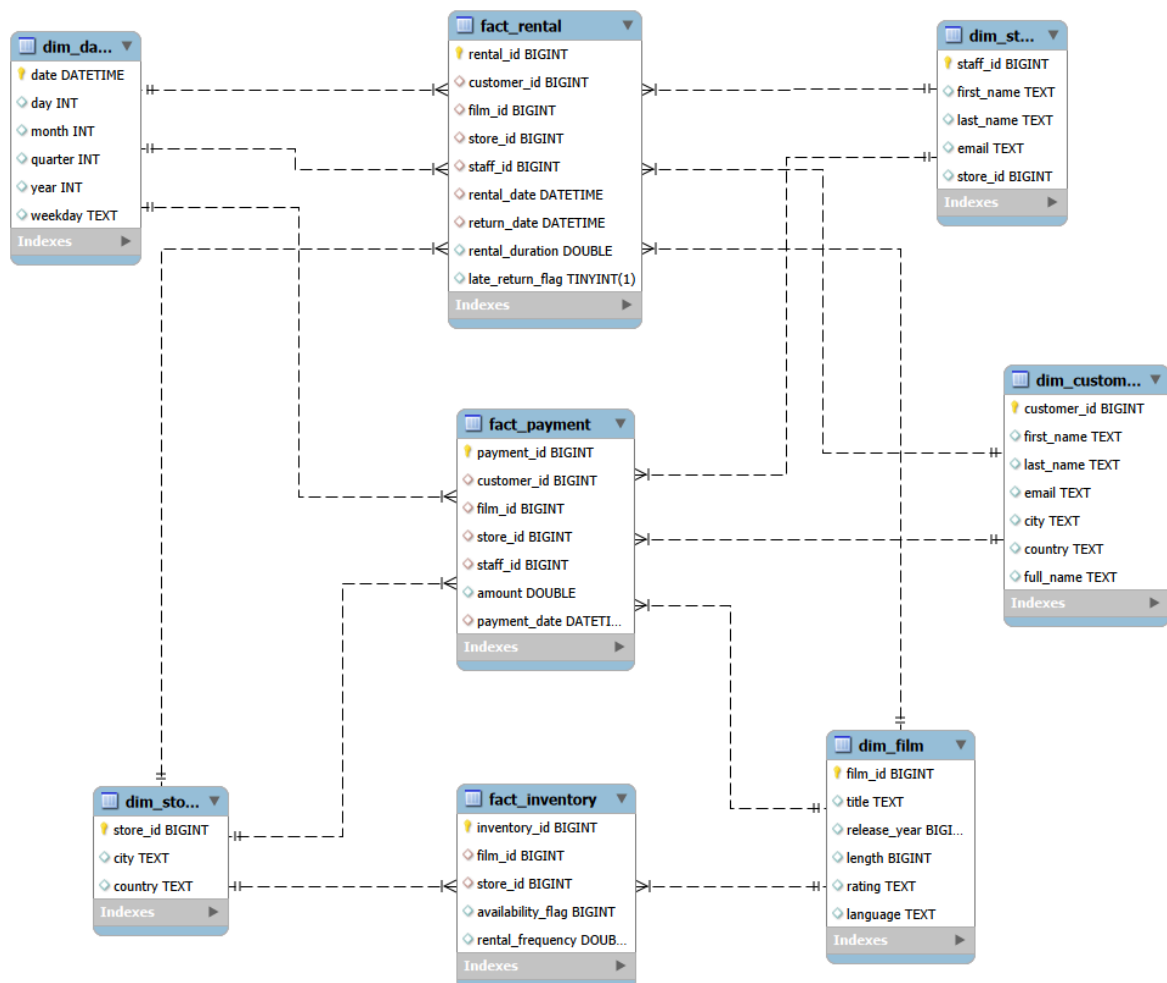
DimDate

Stores calendar attributes such as day, month, quarter, year, and weekday.

3. Dimensional Model Diagram

The warehouse uses a fact constellation schema with three fact tables connected to shared dimensions.

- FactRental connects to Customer, Film, Store, Staff, and Date dimensions.
- FactPayment connects to Customer, Film, Store, Staff, and Date dimensions.
- FactInventory connects to Film and Store dimensions.



4. ETL Design

4.1 Extract

Data is extracted from the following OLTP tables:

- rental
- payment
- customer
- film
- inventory
- store
- staff
- address
- city
- country
- language

4.2 Transform

The following transformations are applied:

- Join related tables to create dimensions.
- Combine first and last names into full names.
- Calculate rental duration.
- Detect late returns.
- Calculate rental frequency.
- Build the date dimension.
- Standardize city, country, and language values.

4.3 Load

Dimension tables are loaded first, followed by fact tables.

Dimensions:

- dim_customer
- dim_film
- dim_store
- dim_staff
- dim_date

Facts:

- fact_rental
- fact_payment
- fact_inventory

5. Business Questions Answered

Q1: Which films are rented most frequently?

```
SELECT f.title, COUNT(r.rental_id) AS rental_count
```

```
FROM fact_rental r
```

```
JOIN dim_film f ON r.film_id = f.film_id
```

```
GROUP BY f.title
```

```
ORDER BY rental_count DESC
```

```
LIMIT 10;
```

Q2: Which stores generate the highest revenue?

```
SELECT s.store_id, s.city, s.country,
```

```
       SUM(p.amount) AS total_revenue
```

```
FROM fact_payment p
```

```
JOIN dim_staff st ON p.staff_id = st.staff_id
```

```
JOIN dim_store s ON st.store_id = s.store_id
```

```
GROUP BY s.store_id, s.city, s.country
```

```
ORDER BY total_revenue DESC;
```

Q3: Which customers generate the highest revenue?

```
SELECT c.full_name, SUM(p.amount) AS lifetime_value
```

```
FROM fact_payment p
```

```
JOIN dim_customer c
ON p.customer_id = c.customer_id
GROUP BY c.customer_id, c.full_name
ORDER BY lifetime_value DESC
LIMIT 10;
```

Q4: Which staff members process the highest number of payments?

```
SELECT s.staff_id, s.first_name, s.last_name,
       COUNT(p.payment_id) AS payments_processed
FROM fact_payment p
JOIN dim_staff s
ON p.staff_id = s.staff_id
GROUP BY s.staff_id, s.first_name, s.last_name
ORDER BY payments_processed DESC;
```

Q5: Which films are returned late most often?

```
SELECT f.title, COUNT(*) AS late_returns
FROM fact_rental r
JOIN dim_film f
ON r.film_id = f.film_id
WHERE r.late_return_flag = TRUE
GROUP BY f.title
ORDER BY late_returns DESC;
```

6. Data Quality Considerations

- Remove duplicate records.
- Validate rental and payment dates.
- Detect missing payment amounts.
- Standardize city and country names.
- Handle null values properly.

7. Conclusion

The dimensional model reorganizes the Sakila OLTP database into a structure optimized for analytics and reporting. The warehouse supports analysis of rentals, payments, inventory, customer activity, and store performance through fact and dimension tables combined with ETL processing.